

Database Relationships

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What is a Relationship?

Definition: A relationship is an association or connection between two or more entities that describes how they interact with each other.

Entity	Relationship
Represents an object	Represents an association
Usually a noun	Usually a verb
Has attributes	Can also have attributes
Rectangle (Chen notation)	Diamond (Chen notation)

Example: Student (Entity), Enrolls (Relationship).

Characteristics of Relationships

- Connects two or more entities.
- Has a name.
- May have attributes.
- Has a degree (Unary, Binary, etc.).
- Has cardinality constraints (covered in the next chapter).
- Has participation constraints (covered later).

Relationship Set

Definition: A relationship set is the collection of all relationships of the same type.

Degree of Relationship

The degree of a relationship tells us: How many entity types participate in the relationship.

Unary Relationship

Definition: A relationship involving one entity type. The entity relates to itself.

Example: Employee manages Employee.

Binary Relationship

Definition: A relationship involving two entity types. This is the most common relationship in databases.

Ternary Relationship

Definition: A relationship involving three entity types simultaneously.

Example: Doctor prescribes Medicine to Patient.

Recursive Relationship

A recursive relationship is a relationship where an entity relates to another instance of the same entity type. This is essentially a special case of a unary relationship. Every recursive relationship is unary because only one entity type participates. Use relationship attributes only when they describe the association itself.

Q6. Why can't every ternary relationship be replaced by binary relationships?

Answer: Because a ternary relationship captures a single association involving three entities simultaneously. Splitting it into binary relationships may lose important business meaning.